Conductors $\frac{\varepsilon(\omega)}{\varepsilon_0} = \frac{\varepsilon_b(\omega)}{\varepsilon_0} + \frac{i\sigma(\omega)}{\varepsilon_0\omega} \approx 1 - \frac{\omega_p^2}{\omega(\omega+\frac{i}{\tau})} = 1 - \frac{\omega_p^2}{\omega^2(1+i\gamma)}$

$$\sigma(\omega) = \frac{\sigma_0}{1-i\omega\tau}, \quad \sigma_0 = \frac{e^2 n\tau}{m} \qquad \gamma \equiv \frac{1}{\omega\tau}$$

$$\sigma(\omega) = \frac{i\varepsilon_0\omega_p^2}{\omega+\frac{i}{\tau}}$$

For $\omega \ll \omega_p$, but arbitrary $\omega\tau$, $\frac{\varepsilon(\omega)}{\varepsilon_0} \approx -\frac{\omega_p^2}{\omega(1+i\gamma)}$

$$\Rightarrow n(\omega) = \frac{\omega_p}{\omega}\frac{1}{\sqrt{-1-i\gamma}} \qquad \left(\text{neglecting } \frac{\varepsilon_b}{\varepsilon_0} \sim 1\right)$$

Need to choose solution which decays inside the conductor $\Rightarrow$ $n''(\omega) > 0$

$$\frac{1}{\sqrt{-1-i\gamma}} = \sqrt{\frac{-1+i\gamma}{(-1-i\gamma)(-1+i\gamma)}} = \sqrt{\frac{-1+i\gamma}{1+\gamma^2}} = \frac{e^{i\varphi/2}}{(1+\gamma^2)^{1/4}}$$

$$-1+i\gamma = (1+\gamma^2)^{1/2}e^{i\varphi} \Rightarrow \begin{cases}(1+\gamma^2)^{1/2}\cos\varphi = -1\\ (1+\gamma^2)^{1/2}\sin\varphi = \gamma\end{cases}$$

$$\tan\varphi = -\gamma$$

$$\varphi = -\tan^{-1}(\gamma) + \pi k$$

$$n''(\omega) \sim \mathrm{Im}\left(e^{+i\varphi/2}\right) = \mathrm{Im}\left(e^{-\frac{i}{2}\tan^{-1}(\gamma)+\frac{i}{2}\pi k}\right)$$

Suppose $\gamma \ll 1 \Rightarrow \tan^{-1}(\gamma) \approx \gamma$

$$\mathrm{Im}\left(e^{-\frac{i}{2}\gamma}\right) = -\sin\frac{\gamma}{2} = -\frac{\gamma}{2} < 0$$

but $\mathrm{Im}\left(e^{i\frac{\pi}{2}-i\frac{\gamma}{2}}\right) = \sin\left(\frac{\pi}{2}-\frac{\gamma}{2}\right) = \cos\frac{\gamma}{2} > 0$

Therefore we choose $\varphi = \pi - \tan^{-1}(\gamma)$

Thus $n(\omega) = \frac{\omega_p}{\omega} \frac{e^{i\frac{\pi}{2} - \frac{i}{2}\tan^{-1}(\gamma)}}{(1+\gamma^2)^{1/4}} = n'(\omega) + i\, n''(\omega)$

$$n''(\omega) = \frac{\omega_p}{\omega} \frac{\sin\left[\frac{\pi}{2} - \frac{1}{2}\tan^{-1}(\gamma)\right]}{(1+\gamma^2)^{1/4}} = \frac{\omega_p \cos\left(\frac{1}{2}\tan^{-1}(\gamma)\right)}{(1+\gamma^2)^{1/4}\omega}$$

$$n'(\omega) = \frac{\omega_p}{\omega(1+\gamma^2)^{1/4}} \cos\left(\frac{\pi}{2} - \frac{1}{2}\tan^{-1}(\gamma)\right) = \frac{\omega_p \sin\left(\frac{1}{2}\tan^{-1}(\gamma)\right)}{(1+\gamma^2)^{1/4}\omega}$$

<u>Limits:</u> (a) $\gamma = \frac{1}{\omega\tau} \ll 1$ (so that $\omega\tau \gg 1$) $\Rightarrow \varepsilon''$ very small.
but $\omega \ll \omega_p$

$$\Rightarrow \tan^{-1}(\gamma) \to \gamma \ll 1$$

$$n''(\omega) \sim \frac{\omega_p}{\omega} \gg 1$$

$$n'(\omega) \sim \frac{\omega_p}{\omega} \sin\left(\frac{\gamma}{2}\right) \doteq \left(\frac{1}{2}\gamma\right)\left(\frac{\omega_p}{\omega}\right) \ll n''(\omega)$$

(b) $\gamma = \frac{1}{\omega\tau} \gg 1 \Rightarrow \omega\tau \ll 1 \Rightarrow \arctan(\gamma) \to \frac{\pi}{2}$

$$\cos\left[\frac{1}{2}\arctan(\gamma)\right] = \frac{1}{\sqrt{2}}$$

$$\sin\left[\frac{1}{2}\arctan(\gamma)\right] = \frac{1}{\sqrt{2}}$$

$$\Rightarrow n(\omega) = \frac{\omega_p}{\sqrt{\gamma}\,\omega} e^{i\frac{\pi}{2} - i\frac{\pi}{4}} = \frac{\omega_p e^{i\pi/4}}{\sqrt{\gamma}\,\omega}$$

$\Rightarrow$ here $n'(\omega) = n''(\omega) = \frac{\omega_p}{\sqrt{2\gamma}\,\omega} \gg 1$

Since $n_2 \gg 1$ we have $n_1 \sin\theta_i = n_2 \sin\theta_T$

$\Rightarrow \sin\theta_T = \frac{n_1 \sin\theta_i}{n_2} \ll 1 \Rightarrow$ refracted wave propagates almost along the normal, $\theta_T \ll 1$.

Consider <u>normal incidence</u> $\theta_I = \theta_T = 0$:

$$R_s = R_p = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 = \left|\frac{n_2' + i n_2'' - n_1}{n_2' + i n_2'' + n_1}\right|^2 =$$

$$= \frac{(n_2' - n_1)^2 + (n_2'')^2}{(n_2' + n_1)^2 + (n_2'')^2} \approx \frac{(n_2')^2\left(1 - \frac{n_1}{n_2'}\right)^2 + (n_2')^2}{(n_2')^2\left(1 + \frac{n_1}{n_2'}\right)^2 + (n_2')^2} =$$

$n_2' = n_2''$

$$= \frac{1 + \left(1 - \frac{n_1}{n_2'}\right)^2}{1 + \left(1 + \frac{n_1}{n_2'}\right)^2} = \frac{1 + 1 - 2n_1/n_2'}{2 + 2n_1/n_2'} = \frac{1 - n_1/n_2'}{1 + n_1/n_2'}$$

$$= 1 - \frac{2n_1}{n_2'} = 1 - 2n_1 \frac{\omega\sqrt{2\gamma}}{\omega_p} = 1 - \frac{n_1}{\omega_p}\sqrt{\frac{8\omega}{\tau}}$$

$$\omega_p = \sqrt{\frac{e^2 n}{m\varepsilon_0}}, \quad \sigma_0 = \frac{e^2 n \tau}{m} = \varepsilon_0 \omega_p^2 \tau$$

$$\Rightarrow 1 - R = n_1\sqrt{\frac{8\omega}{\omega_p^2 \tau}} = n_1\sqrt{\frac{8\varepsilon_0\omega}{\sigma_0}} = \sqrt{\frac{8\varepsilon_1\omega}{\sigma_0}}$$

Using $n_1 = \sqrt{\frac{\varepsilon_1}{\varepsilon_0}}$

Hagen – Rubens relation: $R(\omega) \approx 1 - \sqrt{\frac{8\varepsilon_1\omega}{\sigma_0}}$

high reflectivity
(makes metals shiny)

Q: We know that for $\omega < \omega_p$ the EM wave does <u>not</u> propagate inside the conductor $\Rightarrow$ why $1 - R \neq 0$?

A: It must be energy lost due to Joule's heating! (homework problem)

$$1 - R = 1 - \frac{(n_2' - n_1)^2 + (n_2'')^2}{(n_2' + n_1)^2 + (n_2'')^2} = \frac{4 n_1 n_2'}{(n_2' + n_1)^2 + (n_2'')^2} \approx \frac{4 n_1 n_2'}{(n_2')^2 + (n_2'')^2} =$$

$$= 2n_1/n_2' \;!!!$$

Magneto-optic Kerr ~~rotation~~ effect (MOKE)

Medium = conductor in magnetic field along $\hat{z}$ axis, $\vec{B} = B_0 \hat{z}$.
From Lecture 14 "Faraday rotation" we know
that $\varepsilon_{ij}(\omega) = \varepsilon_0 \left(1 - \frac{\omega_p^2}{\omega^2}\right) \delta_{ij} + i \varepsilon_0 \frac{\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}$

$\Omega_c = \frac{q B_0}{m} \left(= -\frac{e B_0}{m}\right)$

For transverse wave $\vec{E}(z,t) = (E_x, E_y, 0)\, e^{-i\omega t}$

$$\nabla^2 \vec{E} = +\mu \frac{\partial^2 \vec{D}}{\partial t^2} = -\mu \omega^2 \vec{D}$$

[This obtains from $\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\frac{\partial}{\partial t} \vec{\nabla} \times \vec{B} = -\mu \frac{\partial}{\partial t} \underbrace{(\vec{\nabla} \times \vec{H})}_{\frac{\partial \vec{D}}{\partial t}}$

$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\nabla^2 \vec{E} + \vec{\nabla}(\vec{\nabla} \cdot \vec{E})$ $\qquad \vec{B} = \mu \vec{H}$ $\qquad = -\mu \frac{\partial^2 \vec{D}}{\partial t^2}$]

$$D_i = \varepsilon_{ij} E_j$$

$$\begin{cases} \frac{d^2}{dz^2} E_x = -\mu \omega^2 (\varepsilon_{xx} E_x + \varepsilon_{xy} E_y) = -\mu \varepsilon_0 \frac{\omega^2}{c^2 \varepsilon_0 \mu_0} (\varepsilon_{xx} E_x + \varepsilon_{xy} E_y) \\ \frac{d^2}{dz^2} E_y = -\mu \omega^2 (\varepsilon_{yy} E_y + \varepsilon_{yx} E_x) \end{cases}$$

$$\varepsilon_{ij} = \varepsilon_0 \begin{pmatrix} 1 - \frac{\omega_p^2}{\omega^2} & i \frac{\omega_p^2 \Omega_c}{\omega^3} \\ -i \frac{\omega_p^2 \Omega_c}{\omega^3} & 1 - \frac{\omega_p^2}{\omega^2} \end{pmatrix} = \varepsilon_0 \tilde{\varepsilon}_{ij}$$ and let $\mu = \mu_0$

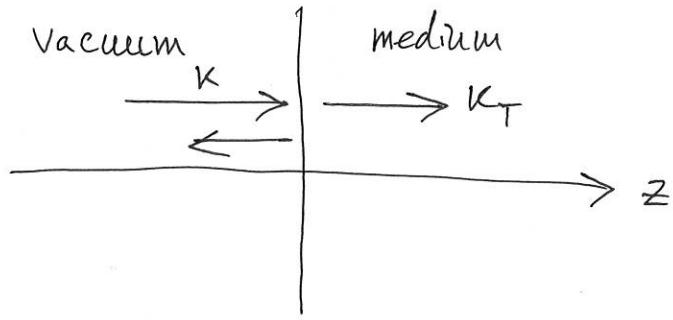
$$\begin{cases} \frac{d^2 E_x}{dz^2} = -\frac{\omega^2}{c^2} (\tilde{\varepsilon}_{xx} E_x + \tilde{\varepsilon}_{xy} E_y) \\ \frac{d^2 E_y}{dz^2} = -\frac{\omega^2}{c^2} (\tilde{\varepsilon}_{yy} E_y - \tilde{\varepsilon}_{xy} E_x) \end{cases}$$

for $z > 0$
$\varepsilon_{yy} = \varepsilon_{xx}$
$\varepsilon_{xy} = -\varepsilon_{yx}$

Transverse wave $\vec{E}(z) = (E_x, E_y, 0)\, e^{i k_T z}$ $\quad z>0$ in medium

Outside the medium $\vec{E}(z) = (E_i, 0, 0)\, e^{ikz} + (E_{Rx}, E_{Ry}, 0) \cdot e^{-ikz}$

$z<0$ $\quad$ incoming $\quad$ reflected



For simplicity, normal incidence $\theta_i = \theta_R = \theta_T = 0$.

For $z>0$:

$$\left(k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2}\right) E_x = -\frac{\omega^2}{c^2}\, \tilde{\varepsilon}_{xy}\, E_y$$

$$\left(k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2}\right) E_y = \frac{\omega^2 \tilde{\varepsilon}_{xy}}{c^2}\, E_x$$

$$\Rightarrow \left(k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2}\right) E_x = -\frac{\left(\frac{\omega^2 \tilde{\varepsilon}_{xy}}{c^2}\right)^2}{k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2}}\, E_x$$

$$\left(k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2}\right)^2 + \left(\frac{\omega^2 |\tilde{\varepsilon}_{xy}|}{c^2}\right)^2 = 0 \qquad \text{Note } \tilde{\varepsilon}_{xy} = i\,\frac{\omega_p^2 \Omega_c}{\omega^3}$$

~~Solve~~ $\cancel{\left(\frac{\omega^2}{c^2}\right)^2 \left(\tilde{\varepsilon}_{xx}^2 + \tilde{\varepsilon}_{xy}^2\right)}$

Treat $\tilde{\varepsilon}_{xy}$ as small correction and solve for k_T by iterations

$$k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2} = \pm \frac{\omega^2 |\tilde{\varepsilon}_{xy}|}{c^2} \qquad \text{Note } \tilde{\varepsilon}_{xx} = \frac{\varepsilon_{xx}}{\varepsilon_0} = n^2$$

$$k_T = \sqrt{\frac{\omega^2 n^2}{c^2} \pm \frac{\omega^2 |\tilde{\varepsilon}_{xy}|}{c^2}} = \frac{\omega n}{c}\left(1 \pm \frac{|\tilde{\varepsilon}_{xy}|}{2\tilde{\varepsilon}_{xx}}\right)$$

Two solutions : $k_T = k_+ = \frac{\omega n}{c} + \frac{\omega |\tilde{\varepsilon}_{xy}|}{2cn}$

$$k_T = k_- = \frac{\omega n}{c} - \frac{\omega |\tilde{\varepsilon}_{xy}|}{2cn}$$

Eigenvectors: (are circularly polarized waves)

$$k_T = k_+ \Rightarrow k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}}{c^2} = \frac{\omega^2 |\tilde{\varepsilon}_{xy}|}{c^2}$$

$$\Rightarrow \quad \frac{\omega^2}{c^2} |\tilde{\varepsilon}_{xy}| E_x = -\frac{\omega^2}{c^2} i |\tilde{\varepsilon}_{xy}| E_y$$

$$E_x = -i E_y$$

$$k_T = k_- \Rightarrow k_T^2 - \frac{\omega^2 \tilde{\varepsilon}_{xx}^2}{c^2} = -\frac{\omega^2}{c^2} |\tilde{\varepsilon}_{xy}|$$

$$\Rightarrow E_x = i E_y$$

$$\Rightarrow \vec{E}(z) = E_+ (1, i, 0) e^{i k_+ z} + E_- (1, -i, 0) e^{i k_- z}$$

$$\Rightarrow \begin{cases} \vec{E}(z=+0) = (E_+ + E_-)\hat{x} + i(E_+ - E_-)\hat{y} \\ \vec{E}(z=-0) = (E_i + E_{Rx})\hat{x} + E_{Ry}\hat{y} \end{cases}$$

$$\Rightarrow \left.\begin{cases} E_i + E_{Rx} = E_+ + E_- \\ E_{Ry} = i(E_+ - E_-) \end{cases}\right\} \#1$$

We also have $\vec{H}_{\parallel}(z=-0) = \vec{H}_{\parallel}(z=+0)$

and for non-magnetic medium $\mu = \mu_0$ this means

$$\vec{B}_{\parallel}\big|_{z=-0} = \vec{B}_{\parallel}\big|_{z=+0}$$

But $i\omega \vec{B} = \vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & 0 \end{vmatrix} = \hat{x}\left(-\frac{\partial E_y}{\partial z}\right)$

$$-\hat{y}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial z}\right)$$

$$+\hat{z}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)$$

$$\Rightarrow i\omega \vec{B} = -\frac{\partial E_y}{\partial z}\hat{x} + \frac{\partial E_x}{\partial z}\hat{y}$$

$\Rightarrow$ two more equations: $\left.\frac{\partial E_{y,x}}{\partial z}\right|_{z=-0} = \left.\frac{\partial E_{y,x}}{\partial z}\right|_{z=+0}$

$$\left.\frac{\partial}{\partial z} E_x\right|_{z=-0} = ikE_i - ikE_{Rx}$$

$$\left.\frac{\partial E_y}{\partial z}\right|_{z=-0} = -ikE_{Ry}$$

$$\left.\frac{\partial}{\partial z} E_x\right|_{z=+0} = E_+ ik_+ + E_- ik_- = i(k_+E_+ + k_-E_-)$$

$$\left.\frac{\partial}{\partial z} E_y\right|_{z=+0} = ik_+(iE_+) + ik_-(-iE_-) =$$

$$= -k_+E_+ + k_-E_-$$

$$\Rightarrow \left.\begin{cases} k(E_i - E_{Rx}) = k_+E_+ + k_-E_- \\ kE_{Ry} = i(-k_+E_+ + k_-E_-) \end{cases}\right\} \#2$$

Let $k = \frac{\omega}{c}$, $k_\pm = nk(1 \pm \Delta)$:

$$\Delta = \frac{|\tilde{\varepsilon}_{xy}|}{2n^2} = \frac{|\tilde{\varepsilon}_{xy}|}{2\tilde{\varepsilon}_{xx}} \ll 1$$

$$\begin{cases} k(E_i - E_{Rx}) = nk\left(E_+ + E_- + \Delta(E_+ - E_-)\right) \\ k\,E_{Ry} = -i\,nk\left((1+\Delta)E_+ - (1-\Delta)E_-\right) = \end{cases}$$

$$= -i\,nk\left(E_+ - E_- + \Delta(E_+ + E_-)\right)$$

Use system (#1):

$$\begin{cases} E_i - E_{Rx} = n\left(E_i + E_{Rx} - i\Delta E_{Ry}\right) \\ E_{Ry} = -n\left(E_{Ry} + i\Delta(E_i + E_{Rx})\right) \end{cases}$$

$$\begin{cases} (1+n)\,E_{Ry} = -in\Delta\,(E_i + E_{Rx}) \\ (1-n)\,E_i = (1+n)\,E_{Rx} - in\Delta\,E_{Ry} \end{cases}$$

$$(1-n)\,E_i = (1+n)\,E_{Rx} - \frac{in\Delta(-in\Delta)}{1+n}(E_i + E_{Rx})$$

$$= (1+n)\,E_{Rx} - \underbrace{\frac{n^2\Delta^2}{1+n}(E_i + E_{Rx})}_{\text{neglect } \Delta^2 \text{ terms!}}$$

$$\Rightarrow E_{Rx} \approx \frac{1-n}{1+n}\,E_i$$

$$\Rightarrow E_{Rx} + E_i = \left(1 + \frac{1-n}{1+n}\right)E_i = \frac{2\,E_i}{1+n}$$

$$\Rightarrow E_{Ry} = \frac{-in\Delta}{1+n}\,\frac{2}{1+n}\,E_i = \frac{-i\,2n\Delta}{(1+n)^2}\,\frac{1+n}{1-n}\,E_{Rx}$$

$$E_{Ry} = -\frac{i\,2n\Delta}{1-n^2}\,E_{Rx} = -\frac{i\,|\tilde{\varepsilon}_{xy}|}{n(1-n^2)}\,E_{Rx}$$

$$2n\Delta = \frac{|\tilde{\varepsilon}_{xy}|}{n}$$

Thus, the reflected wave reads $(z<0)$

$$\vec{E}_R(z,t) = (E_{Rx}, E_{Ry}, 0)\, e^{-ikz}\, e^{-i\omega t} =$$

$$= \underset{\substack{\| \\ \frac{1-n}{1+n} E_i}}{E_{Rx}} \left(1, \frac{i|\tilde{\varepsilon}_{xy}|}{n(n^2-1)}, 0\right) e^{-ikz-i\omega t}$$

$$n^2 = \frac{\varepsilon_{xx}}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2} \Rightarrow \begin{cases} n = \sqrt{1-\frac{\omega_p^2}{\omega^2}} & \text{for } \omega > \omega_p \\ n = i\sqrt{\frac{\omega_p^2}{\omega^2}-1} & \text{for } \omega < \omega_p \end{cases}$$

$\boxed{\omega<\omega_p:}$ In the conductor $e^{ik_T z}$ $\left(k_T = \frac{\omega}{c} n\right) \Rightarrow$ for $\omega<\omega_p$ we have $e^{iz\frac{\omega}{c} i\left(\frac{\omega_p^2}{\omega^2}-1\right)^{1/2}} = \exp\left(-\frac{z}{c}\sqrt{\omega_p^2-\omega^2}\right)$ decays exponentially $(z>0) \Rightarrow$ full reflection.

$$\Rightarrow \text{for } \omega<\omega_p \quad E_{Ry} = \frac{i|\tilde{\varepsilon}_{xy}|\, E_{Rx}}{i\sqrt{\frac{\omega_p^2}{\omega^2}-1}\left(-\frac{\omega_p^2}{\omega^2}\right)} = -\frac{\omega^3|\tilde{\varepsilon}_{xy}|}{\sqrt{\omega_p^2-\omega^2}\,\omega_p^2} E_{Rx} =$$

$$= -\frac{\cancel{\omega_p^2}\,\Omega_c}{\sqrt{\omega_p^2-\omega^2}} E_{Rx}$$

$$\Rightarrow \vec{E}_R(z,t) = \mathrm{Re}\left\{E_{Rx}\left(\hat{x} - \frac{\Omega_c}{\sqrt{\omega_p^2-\omega^2}}\hat{y}\right) e^{-ikz-i\omega t}\right\}$$

$$= \left(\hat{x} - \frac{\Omega_c}{\sqrt{\omega_p^2-\omega^2}}\hat{y}\right) E_{Rx} \cos(kz+\omega t)$$

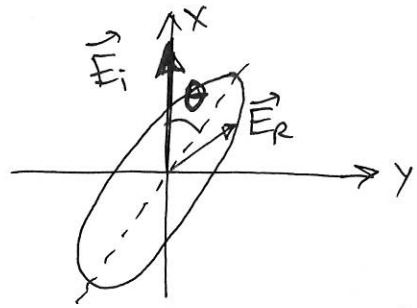
describes linearly polarized EM wave.
Its polarization is rotated from (toward $-\hat{y}$ axis) relative to the polarization of the incipient wave $\vec{E}_i$ (which is along $\hat{x}$).

$\boxed{\omega > \omega_p:}$ $n = \sqrt{1 - \frac{\omega_p^2}{\omega^2}}$ is real, EM wave propagates in the conductor.

Reflected wave is now <u>elliptically</u> polarized

$$\vec{E}_R(z,t) = \mathrm{Re}\left\{ E_{Rx}\left(\hat{x} + \frac{i\,|\tilde{\varepsilon}_{xy}|}{n(n^2-1)}\,\hat{y}\right) e^{-ikz-i\omega t} \right\}$$

$$= E_{Rx}\cos(kz+\omega t)\,\hat{x} + E_{Rx}\,\frac{|\tilde{\varepsilon}_{xy}|}{n(n^2-1)}\sin(kz+\omega t)\,\hat{y}.$$



Its main polarization axis has rotated by

$$\theta = -\frac{\omega^3}{\sqrt{\omega^2-\omega_p^2}\,(\omega_p^2)} \cdot \frac{\omega_p^2\,\Omega_c}{\omega^3} = -\frac{\Omega_c}{\sqrt{\omega^2-\omega_p^2}}.$$